

TABLE 6—LINEAR-IN-MEANS MODELS OF PEER EFFECTS IN ACHIEVEMENT FOR LOUISIANA
(OLS and 2SLS models using evacuee share in grade $\times$ mean evacuee lagged achievement
as an instrument for peer achievement)

	OLS— Pre-Katrina only	OLS— all years	2SLS— first stage	2SLS— second stage	OLS model with peer achievement interacted with own quartile			
	Avg. peer score (1)	Avg. peer score (2)	Evac. share $\times$ mean lagged evac. score (3)	Avg. peer score (4)	Avg. peer score $\times$ quartile 1	Avg. peer score $\times$ quartile 2	Avg. peer score $\times$ quartile 3	Avg. peer score $\times$ quartile 4 (5)
<i>Panel A. Elementary (no lags)</i>								
Math	0.56*** (0.02)	0.58*** (0.02)	1.69*** (0.38)	0.33** (0.15)	0.29*** (0.02)	0.32*** (0.02)	0.33*** (0.02)	0.40*** (0.02)
Observations	251,763	287,090	287,090	287,090		287,090		
F-test of joint equality						5.2**		
ELA	0.44*** (0.03)	0.46*** (0.03)	1.07*** (0.36)	−0.00 (0.27)	0.29*** (0.03)	0.25*** (0.02)	0.24*** (0.03)	0.25*** (0.02)
Observations	251,805	287,139	287,139	287,139		287,139		
F-test of joint equality						0.9		
<i>Panel B. Middle/High (with lags)</i>								
Math	0.31*** (0.04)	0.34*** (0.06)	2.55*** (0.82)	0.15* (0.08)	0.27*** (0.03)	0.29*** (0.04)	0.26*** (0.09)	0.41*** (0.14)
Observations	192,843	535,302	535,302	535,302		535,302		
F-test of joint equality						3.7**		
ELA	0.25*** (0.02)	0.41*** (0.02)	1.91*** (0.43)	0.08 (0.08)	0.42*** (0.02)	0.41*** (0.02)	0.32*** (0.02)	0.28*** (0.04)
Observations	179,217	520,569	520,569	520,569		520,569		
F-test of joint equality						13.9***		

(Continued)

using school-level shares. For elementary school the results are similar to Table 5. For middle/high school the results remain negative and are bigger in magnitude, as in Houston, with 9 out of 16 coefficients significant.

C. Linear-in-Means Models of Peer Effects in Louisiana

After estimating reduced-form regressions, we proceed to estimate equation (2) for elementary and middle/high school students in Louisiana. Columns 1 and 2 in Table 6 report the OLS estimates for the pre-Katrina and full samples. Columns 3 and 4 report the first and second stages of our instrumental variables specification. We use the achievement-adjusted evacuee share in the school and grade, i.e., the evacuee share multiplied by the mean pre-Katrina achievement of the evacuees, to instrument for mean peer achievement. Thus, the identification assumption here is

TABLE 6—LINEAR-IN-MEANS MODELS OF PEER EFFECTS IN ACHIEVEMENT FOR LOUISIANA
(*OLS and 2SLS models using evacuee share in grade $\times$ mean evacuee lagged achievement
as an instrument for peer achievement*) (Continued)

	Reduced-form model with peer achievement interacted with own quartile				2SLS model with peer achievement interacted with own quartile—second stage; instrumented with evacuee share $\times$ mean lagged evacuee score $\times$ own quartile			
	Avg. evacuee score $\times$ evacuee share $\times$ quartile 1	Avg. evacuee score $\times$ evacuee share $\times$ quartile 2	Avg. evacuee score $\times$ evacuee share $\times$ quartile 3	Avg. evacuee score $\times$ evacuee share $\times$ quartile 4	Avg. peer score $\times$ quartile 1	Avg. peer score $\times$ quartile 2	Avg. peer score $\times$ quartile 3	Avg. peer score $\times$ quartile 4
	(6)				(7)			
<i>Panel A. Elementary (no lags)</i>								
Math	1.14** (0.47)	0.96*** (0.34)	1.99*** (0.42)	1.57*** (0.55)	0.80*** (0.27)	0.62*** (0.16)	1.00*** (0.15)	0.76*** (0.19)
Observations	287,090				287,090			
F-test of joint equality	2.7**				1.4			
ELA	1.12* (0.60)	−0.07 (0.31)	0.81** (0.41)	0.49 (0.33)	0.83** (0.36)	−0.09 (0.39)	0.72*** (0.25)	0.47 (0.29)
Observations	287,139				287,139			
F-test of joint equality	2.3*				1.3			
<i>Panel B. Middle/High (with lags)</i>								
Math	0.69*** (0.23)	1.19*** (0.21)	0.77*** (0.22)	0.88*** (0.31)	0.35*** (0.10)	0.64*** (0.22)	0.23* (0.13)	0.31 (0.21)
Observations	535,302				535,302			
F-test of joint equality	2.9**				4.6***			
ELA	0.61** (0.27)	0.42* (0.23)	1.00*** (0.21)	0.45* (0.26)	0.38*** (0.12)	0.21* (0.11)	0.58*** (0.18)	0.23 (0.20)
Observations	520,569				520,569			
F-test of joint equality	3.8**				2.6*			

Notes: Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses nonevacuees in schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Placebo regressions are limited to pre-Katrina years and apply the 2006–2007 and 2007–2008 evacuee shares to 2003–2004 and 2004–2005 observations, respectively. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. In regressions 5, 6, and 7 all covariates are interacted with the student's quartile to generate fully interacted models. Middle/High includes grade 10 in 2001–2002 and 2002–2003, grades 8 and 10 in 2003–2004 and 2004–2005, and grades 6–10 in 2005–2006 and 2006–2007.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.